ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

14 October 2026 · Arjay Arnao, AI231 ME2 · Model · Dataset · Training on A100 · Validation on Raspberry Pi 5

Model

Mic · 16 kHz → log-mel 64 × ≤501 (10 ms hop, ≤ 5 s)

Wake word "Hello / Hi Elkit" · openWakeWord + Elkit head

Conformer encoder × 4 · 96-dim
non-causal · once per command after the wake word (no KV cache)

Intent head (19 + reject) | **Slot head (18 values)**

Actuator {intent, slot} | **Raspberry Pi 5 (8 GB)**

Item	Value
Parameters / weights	intent 0.64 M · 2.7 MB · slot 0.64 M · 2.7 MB · wake head 0.21 M · 0.9 MB (+ frozen 0.6 M front end, 2.4 MB)

Validation on the Raspberry Pi 5

Item	Value
Keyword / intent acc	wake word 86.6 % generated / 65.6 % real-speaker phrases (test, 0.886 × 2) · whole command (intent + slot) 98.39 % holdout, 97.60 % test
False-accept rate	wake word 1.5 % of real sentences, 0 / 6,194 real words · out-of-scope commands accepted 31.6 % (test)
Latency p95 / RTF	36.9 ms / 0.01 per command (p50 18.8 ms) · wake word 70.0 ms per 80 ms step, RTF 0.22
Holdout on the Pi	intent 98.39 % (183/186) · slot 100 % · out-of-scope 5/16 (202 clips)
Runtime	onnxruntime · 1 thr (Pi 5 CPU, 202 holdout clips)

Dataset

Item	Value
Source	ai231 ME2 voice commands (Hugging Face airimonda/ai231-me2-voice-commands @ 5abbe539); licence "other", research/education
Hours / utts	14.8 h / 25,620 training clips (+ 0.73 h / 1,392 validation); test 4,367 · holdout 186
Speakers	246 training speakers/voices · 43 validation · 104 test · 4 holdout (all disjoint)
Labels	19 intents · 18 slots (3 values × 6 slotted intents)

Training on the A100 cluster

Item	Value
Cluster	1 × NVIDIA A100 80GB (uperdfi-a100gpu0), 26.7 min training (job 65672), seed 23102
Objective	cross-entropy, class-weighted, label smoothing 0.1, mixup (utterance classification, not CTC / attention)
Optimiser	AdamW lr 2e-3, wd 0.05, 5-epoch warm-up + cosine, batch 128, bf16, EMA
Steps / loss	24,720 steps (120 epochs, best epoch 112) / train 0.879 · validation 0.262

Live test on the Pi 5 (speaker → mic)

Item	Value
Intent / command	91.1 % / 90.6 % · slot 99.0 % (202 clips)
Errors	wrong 0.5 % · ignored 4.3 % · out-of-scope 9/16
Latency / RTF	21 ms (p95 43 ms) / 0.008 · reply p95 1.21 s

To Be Submitted

GitHub repository	github.com/arjAIareNOW/MLEx-2-VCM- — public, MIT (code)
Dataset location	Hugging Face airimonda/ai231-me2-voice-commands @ 5abbe539a46b9b26ff24e73d2860d1698d44e81f; licence "other" (research/education); DOI 10.57967/hf/10723
A100 cluster	uperdfi-a100gpu0, 1 × A100 80 GB, 29.8 min job (26.7 min training), seeds 23102 (training) / 23101 (split)
Model weights	release/ and deployment/models/ in the repository; research/education use (ai231 terms); Elkit head non-commercial (MMS-TTS data)

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	done: bash reproduce.sh (Slurm) or --local; fresh-clone run checked on the build cluster
2	Dataset licensed and citable (DOI)	done: licensed ("other", research/education), DOI 10.57967/hf/10723, pinned revision + SHA-256
3	Training logs + final checkpoint committed	done: release/intent_conformer/, release/slot_conformer/ (best.pt, ONNX, config, history, log)
4	Pi latency reproduced by the posted script	done: deployment/benchmark_pi.py on the Pi 5 → 36.9 ms p95, RTF 0.01 (1 thread); live class vcm-benchmark: 91.1 % intent, 1.21 s p95 response
5	Held-out test set, unseen speakers	done: test (104 speakers) and holdout (4 speakers) never seen in training or validation
6	Baseline of comparable size compared	done: CNN + BiGRU 405 k params, test 97.41 % vs Conformer 98.01 % (same data and recipe)